



AMD – ITRI PROJECT REVIEW

MI300X Enterprise Document RAG Project Report

Enterprise document retrieval and source-grounded Chinese
answer generation

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AMD Instinct MI300X • PyTorch 2.3 / ROCm 6.2 • Implemented RAG prototype

RAG turns scattered PDF and Excel content into grounded answers

BUSINESS PROBLEM

Enterprise knowledge is spread across PDF reports and Excel sheets. Finding the right information manually is slow, and answers often lack a traceable source.

IMPLEMENTED SOLUTION

Retrieval-Augmented Generation (RAG): the system first retrieves the most relevant document chunks, then Qwen2.5-7B-Instruct generates a Traditional Chinese answer grounded in them.

1 Data Preparation

PDF + Excel → text extraction → 500-char chunks → BGE embeddings → vector index

2 Retrieval

User question → query embedding → dot-product similarity → Top-3 chunks

3 Generation

Retrieved context + sources → Qwen2.5-7B-Instruct → Traditional Chinese answer with filenames

Objective: a searchable knowledge base that answers with sources

PROBLEM ADDRESSED

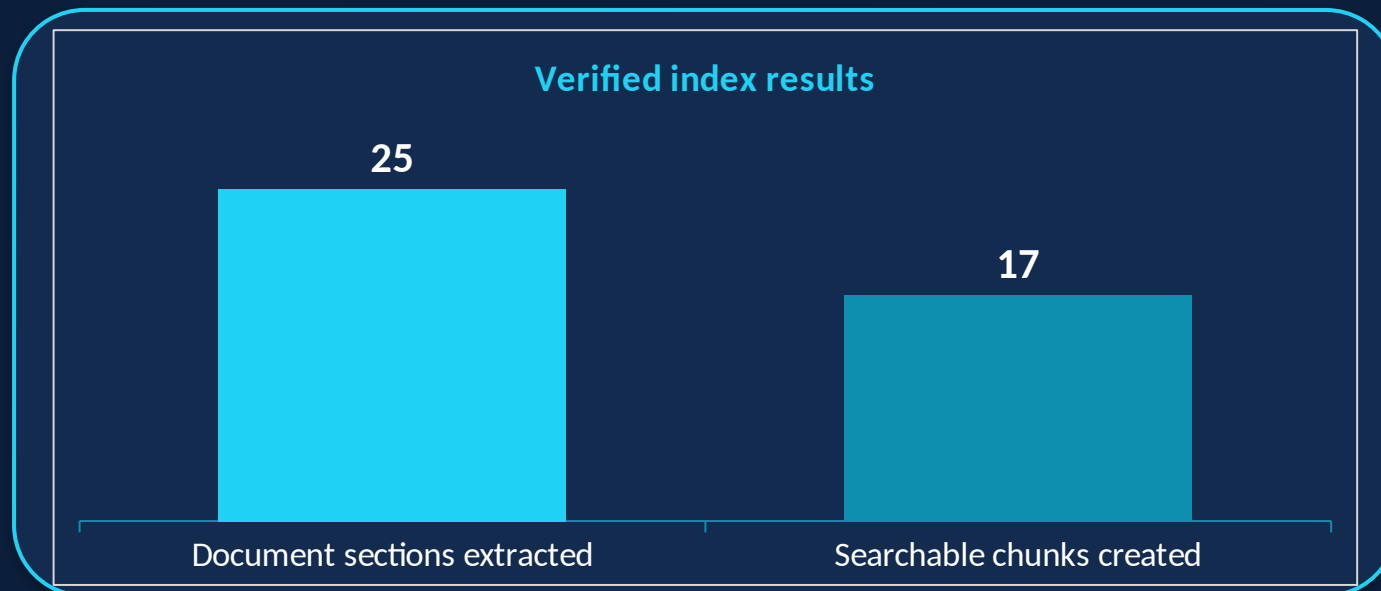
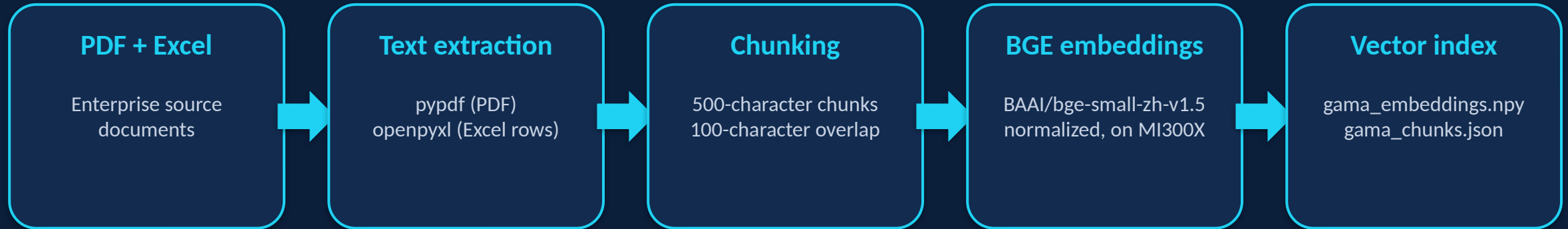
- Information distributed across PDF and Excel files
- Manual search is time-consuming
- Answers may lack document sources



IMPLEMENTED OBJECTIVE

- Build a searchable document knowledge base
- Retrieve relevant chunks
- Generate source-grounded Traditional Chinese answers

Documents become a persistent vector index in five steps

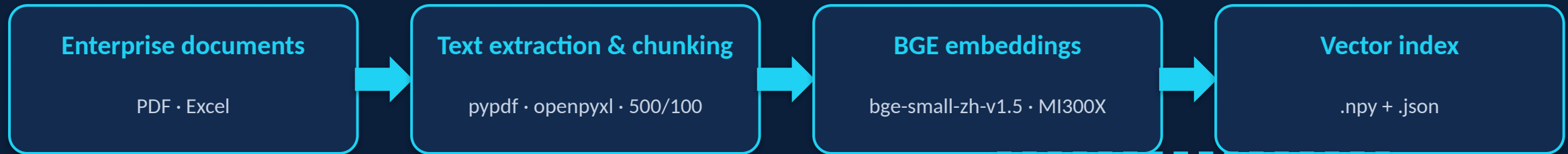


INDEX FACTS

- 25 document sections extracted
- 17 searchable text chunks created
- Embeddings normalized for dot-product similarity
- Index persisted to disk for reuse across queries

The architecture links document evidence to the generated answer

OFFLINE · INDEX CONSTRUCTION



ONLINE · QUESTION ANSWERING



Runs on AMD Instinct MI300X · PyTorch 2.3 · ROCm 6.2

The system retrieves relevant document evidence before Qwen2.5-7B generates the answer.

Two models, two roles: retrieval and generation

BAAI/bge-small-zh-v1.5

Role: Embedding and retrieval

- Converts document chunks and user queries into semantic vectors
- Supports normalized vector search and Top-3 retrieval

+

Qwen2.5-7B-Instruct

Role: Answer generation

- Uses retrieved context and system instructions
- Generates source-grounded Traditional Chinese answers

No model training or fine-tuning is claimed in this project.

Verified results on AMD Instinct MI300X

25

document sections extracted

17

searchable chunks created

2.55 s

observed Qwen response-
generation time

Top-3

retrieved chunks per query

Verified index results

25



Document sections extracted

17



Searchable chunks created

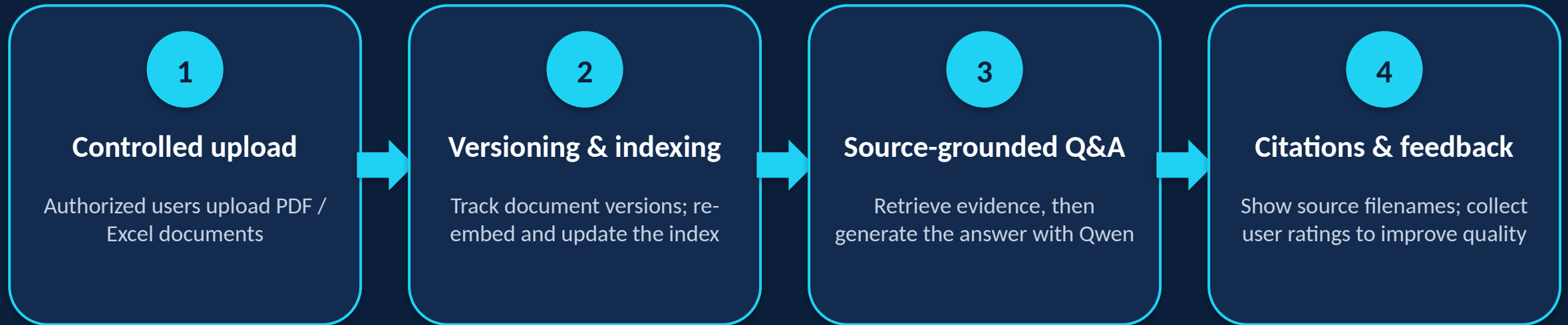
OBSERVATIONS

- Qwen2.5-7B-Instruct loaded and executed successfully on MI300X (bfloat16, device_map=auto)
- The answer used the retrieved documents and stated “insufficient information” instead of inventing a third source

Note: 2.55 seconds is an observed single-query generation time, not a general GPU benchmark.

Proposed deployment model for enterprise RAG

PROPOSED · NOT YET IMPLEMENTED



Future-state interface (concept)

Q: Ask a question about your enterprise documents...

A: Grounded answer · Sources: report.pdf, data.xlsx · Was this helpful?
Yes / No

Proposed roadmap; the current project validated the underlying MI300X RAG workflow.

Request for renewed MI300X access

WHAT THE INITIAL ALLOCATION ACHIEVED

The first MI300X allocation validated the technical feasibility of the full workflow on AMD hardware with PyTorch 2.3 / ROCm 6.2:

- Document indexing (PDF + Excel → 17 chunks)
- Normalized Top-3 semantic retrieval
- Qwen2.5-7B-Instruct inference (bfloat16)
- Source-grounded Traditional Chinese answers

PLANNED NEXT STEPS WITH RENEWED ACCESS

- 1 Expand official product and service documents
- 2 Create a 50–100 question evaluation set
- 3 Measure retrieval quality and inference performance reproducibly
- 4 Build the proposed enterprise knowledge interface

Thank you to AMD and ITRI for enabling this work.